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Computer system, computer, and method for controlling conveyance system

Abstract

A computer system comprising a transportation system and a control system, wherein: the transportation system comprises a plurality of transportation modules having transportation means for transporting articles; the transportation system transmits, for each of the plurality of transportation modules configuring a transportation line, connection information pertaining to connections between transportation modules; and the control system controls the transportation system on the basis of the multiple pieces of connection information.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application incorporates by reference the subject matter of Japanese Patent Application No. 2020-212029, filed on Dec. 22, 2020, and claims priority therefrom.

TECHNICAL FIELD

(2) The present invention relates to control of a conveyance system that conveys articles.

BACKGROUND ART

(3) In a conveyance system, articles such as products, parts, and goods are conveyed using a conveyor line formed by connecting devices such as roller conveyors. Techniques for designing and changing the configuration of a conveyor line according to the expansion and changes of business, changes in demand, and the like is attracting attention. For example, a technique described in PTL 1 is known.

(4) PTL 1 describes “a controller capable of controlling a linear conveying device or a right-angle conveying device that forms a part of a roller conveyor device having a motor. The linear conveying device is a device that conveys a conveyance object in a straight direction and is divided into control zones, each of which includes a motor, and the right-angle conveying device is a device that can change a conveying direction of the conveyance object and includes a plurality of motors. The controller includes: a motor control board, the motor control board includes a plurality of motor drive circuits, a rewritable memory in which a rewritable conveyance program that controls the linear conveying device or the right-angle conveying device is stored, and a CPU; and an input/output circuit that can receive signals from a plurality of detection means provided in the linear conveying device and the right-angle conveying device. The controller can create the conveyance program using a program creation support program. The program creation support program allows inputs of the following items: a name item in which a name that is arbitrarily determined by a user is input as a name for a series of operations formed by combining one or a plurality of operations; a condition item in which a condition under which the series of operations input in the name item are executed is input; an operation item in which a specific content of the series of operations is input; and a next process item in which a name described in the name item corresponding to a series of operations that is executed after the series of operations is input. One condition item and one operation item are processed as a set of condition operation sets. The condition operation set is associated with one or a plurality of next process items and one name item to form a program table”.

(5) By using the technique described in PTL 1, a user can design a conveyor line using a computer for design, and can construct and run the conveyor line by connecting devices on the basis of the design.

CITATION LIST

Patent Literature

(6) PTL 1: JP 2011-230881 A

SUMMARY OF INVENTION

Technical Problem

(7) In the prior art, it is necessary for a user to manually connect devices such as conveyors forming a conveyor line according to the design. Furthermore, in a case where special control logic needs to be set up, the user needs to check the identification information of the device and manually set up the control logic.

(8) In the prior art, in a case where the design of the conveyor line has been changed on the conveyance system side, there is a problem that the change of the conveyor line is not reflected in the control system. In this case, a discrepancy occurs between the conveyor line on the control system side and the conveyor line on the conveyance system side, and there is a possibility that the entire system does not function correctly.

(9) An object of the present invention is to realize a technique of reflecting information regarding a

change in design of a conveyor line constructed in a conveyance system to a control system.
Solution to Problem
(10) A representative example of the invention disclosed in the present application is as follows. That is, a computer system including a conveyance system, and a control system communicably connected to the conveyance system, in which the conveyance system includes a plurality of conveyance modules each having a conveyance means for conveying an article, the control system includes at least one computer having an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, the conveyance system transmits, to the control system, interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line, and the control system controls the conveyance system on the basis of a plurality of pieces of the interconnection information.
Advantageous Effects of Invention
(11) According to the present invention, it is possible to reflect information regarding a change in design of a conveyor line in a conveyance system to a control system. Problems, configurations, and effects other than those described above will be clarified by the following description of embodiments.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a diagram illustrating a configuration example of a system of a first embodiment.
(2) FIG. **2**A is a diagram illustrating an example of a conveyance module in the first embodiment.
(3) FIG. **2**B is a diagram illustrating an example of a conveyance module in the first embodiment.
(4) FIG. **2**C is a diagram illustrating an example of a conveyance module in the first embodiment.
(5) FIG. **3**A is a diagram illustrating a specific structure of the conveyance module in the first embodiment.
(6) FIG. **3**B is a diagram illustrating a specific structure of the conveyance module in the first embodiment.
(7) FIG. **3**C is a diagram illustrating a specific structure of a conveyance module in the first embodiment.
(8) FIG. **4** is a table illustrating an example of a data structure of conveyance module basic information in the first embodiment.
(9) FIG. **5** is a table illustrating an example of a data structure of interconnection information in the first embodiment.
(10) FIG. **6** is a flowchart describing an example of processing executed by an integrated control computer in the first embodiment.
(11) FIG. **7** is a diagram illustrating an example of a conveyor line constructed in a conveyance system in the first embodiment.
(12) FIG. **8** is a diagram illustrating a specific structure of a conveyance module of a second embodiment.
(13) FIG. **9** is a diagram illustrating an example of a conveyor line constructed in a conveyance system in the second embodiment.
(14) FIG. **10** is a diagram illustrating an example of a conveyor line constructed in the conveyance system in the second embodiment.
DESCRIPTION OF EMBODIMENTS
(15) Hereinafter, embodiments of the present invention will be described with reference to the drawings. However, the present invention is not to be construed as being limited to the contents of the description of the embodiments described below. Those skilled in the art can easily understand

that the specific configuration can be changed without departing from the spirit or gist of the present invention.
(16) In the configurations of the invention described below, the same or similar configurations or functions are denoted by the same reference numerals, and redundant description is omitted.
(17) Notations such as “first”, “second”, and “third” in the present specification and the like are used to identify constituent elements, and are not necessarily intended to limit the number or order.
(18) In order to facilitate understanding of the invention, the position, size, shape, range, and the like of each configuration illustrated in the drawings and the like may not represent the actual position, size, shape, range, and the like. Therefore, the present invention is not limited to the position, size, shape, range, and the like disclosed in the drawings and the like.
First Embodiment
(19) FIG. **1** is a diagram illustrating a configuration example of a system of a first embodiment.
(20) The system is formed of an integrated control computer **100** and a conveyance system **101**. The integrated control computer **100** and the conveyance system **101** are connected to each other via a network **102**. The network **102** is a wide area network (WAN), a local area network (LAN), or the like, and a connection method may be either wired or wireless.
(21) The conveyance system **101** is formed of a plurality of conveyance modules **140**. The conveyance system **101** includes a plurality of kinds of conveyance modules **140** having different conveyance means, shapes, and the like. Note that the conveyance system **101** may include an edge device and the like.
(22) In the conveyance system **101**, a conveyor line is constructed using the conveyance modules **140**. Articles are conveyed from a start point to a goal point according to a conveyance route on the conveyor line. Here, the conveyance route represents a conveyance order (conveyance plan) of the articles in the conveyor line.
(23) The conveyance module **140** is a device that performs conveyance work of articles, and includes a controller **150**, a conveyance work device group **151**, a communication device **152**, and an interconnection information acquisition device **153**.
(24) The controller **150** is a device that controls the entire conveyance module **140**, and includes a processor and a memory. The memory stores basic control logic for realizing the basic operation of the conveyance work device group **151**.
(25) The conveyance module **140** automatically conveys articles on the basis of the basic control logic after a conveyor line is constructed. Note that in a case where special control logic is set up from the integrated control computer **100**, the conveyance module **140** conveys the articles on the basis of the special control logic.
(26) Thus, the conveyance module **140** of the present embodiment can autonomously run on the basis of the basic control logic. Therefore, the integrated control computer **100** does not need to set up the control logic (special control logic) for all the conveyance modules **140** forming the conveyor line. Accordingly, there is an advantage that the calculation cost required for setting up the conveyor line can be reduced. Furthermore, there is an advantage that the amount of communication for setting up the control logic can also be reduced.
(27) The conveyance work device group **151** is a device group for performing conveyance work. The conveyance work device group **151** is, for example, a roller, a belt, a sensor, a motor, a lift, an arm, a tire, and the like. Note that the present invention is not limited to the devices included in the conveyance work device group **151**.
(28) The communication device **152** is a device for communicating with other devices.
(29) The interconnection information acquisition device **153** is a device for acquiring interconnection information **130**. Here, the interconnection information **130** is information regarding the connection between the conveyance modules **140** forming the conveyor line. A specific data structure of the interconnection information **130** will be described with reference to FIG. **5**.

(30) The conveyance module **140** in the first embodiment acquires the interconnection information **130** regarding the connection between the conveyance module **140** itself and other conveyance modules **140** using the interconnection information acquisition device **153**. Furthermore, the conveyance module **140** transmits the interconnection information **130** to the integrated control computer **100** using the communication device **152**.
(31) Note that a device included in the conveyance work device group **151** may function as the interconnection information acquisition device **153**.
(32) The integrated control computer **100** is a computer that controls the conveyance system **101**, and includes a processor **110**, a memory **111**, and a communication device **112**. Note that the integrated control computer **100** may include a storage device such as a hard disk drive (HDD) or a solid state drive (SSD), may include an input device such as a keyboard, a mouse, or a touch panel, and may include an output device such as a display.
(33) The processor **110** executes a program stored in the memory **111**. The processor **110** operates as a functional unit that realizes a specific function by executing processing according to the program. In the following description, in cases where processing is described using a functional unit as a subject, it indicates that the processor **110** executes the program that realizes the functional unit. The memory **111** stores the program executed by the processor **110** and information used by the program. Furthermore, the memory **111** is also used as a work area. The communication device **112** is a device for communicating with other devices.
(34) The memory **111** in the first embodiment stores a program that realizes a control unit **120**, and a conveyance module basic information **121**.
(35) The control unit **120** performs multiple kinds of controls such as generation of conveyor line model information for managing a constitution of a conveyor line, generation of a conveyance route, and setting up of control logic.
(36) The conveyance module basic information **121** is information for managing structures and the like of various types of conveyance modules **140**. A specific data structure of the conveyance module basic information **121** will be described with reference to FIG. **4**.
(37) Note that a system that realizes the functions included in the integrated control computer **100** may be configured using a plurality of computers.
(38) FIGS. **2**A, **2**B, and **2**C are diagrams illustrating examples of the conveyance modules **140** in the first embodiment.
(39) A conveyance module **140-1** illustrated in FIG. **2**A is a short straight conveyor. The view (1) in FIG. **2**A is a perspective view of the conveyance module **140-1**, and the view (2) in FIG. **2**A is a top view of the conveyance module **140-1**. Note that axes illustrated in the perspective view and the top view are provided for description.
(40) The conveyance module **140-1** includes a conveyance means that conveys articles from either of two directions (direction A or B) to either of two directions. Note that the directions A and B are uniquely determined with respect to the conveyance module **140-1**.
(41) A conveyance module **140-2** illustrated in FIG. **2**B is a long straight conveyor. The view (1) in FIG. **2**B is a perspective view of the conveyance module **140-2**, and the view (2) in FIG. **2**B is a top view of the conveyance module **140-2**. Note that axes illustrated in the perspective view and the top view are provided for description.
(42) The conveyance module **140-2** includes a conveyance means that conveys articles from either of two directions (direction A or B) to either of two directions. Note that the directions A and B are uniquely determined with respect to the conveyance module **140-2**.
(43) A conveyance module **140-3** illustrated in FIG. **2**C is a diverting conveyor. The view (1) in FIG. **2**C is a perspective view of the conveyance module **140-3**, and the view (2) in FIG. **2**C is a top view of the conveyance module **140-3**. Note that axes illustrated in the perspective view and the top view are provided for description.
(44) The conveyance module **140-3** includes a conveyance means that conveys articles from any of

four directions (directions A, B, C, and D) to any of four directions. Note that the s A, B, C, and D are uniquely determined with respect to the conveyance module **140-3**.

(45) Note that the conveyance modules **140** illustrated in FIGS. **2**A, **2**B, and **2**C are examples, and the present invention is not limited thereto. The conveyance modules **140** also include a conveyor (level changer) that conveys articles up and down, a picking robot that performs picking work, and the like.

(46) FIGS. **3**A, **3**B, and **3**C are diagrams illustrating specific structures of the conveyance modules **140** in the first embodiment.

(47) FIG. **3**A is a perspective view illustrating a specific structure of the conveyance module **140-1** (short straight conveyor). Note that axes illustrated in the perspective view are provided for description.

(48) The conveyance module **140-1** includes rollers **301**, presence sensors **302**, barcode readers **303**, ID markers **304**, a presence indicator LED **305**, a light **306**, and a camera **307**.

(49) The rollers **301** rotate in a constant direction to convey articles. The article may be conveyed while accommodated in a tray. In this case, two or more articles may be accommodated in one tray.

(50) The presence sensors **302** measure the presence or absence of an article on the conveyance module **140-1**. The presence indicator LED **305** outputs a signal indicating the presence or absence of an article on the conveyance module **140-1** on the basis of measurement results of the presence sensors **302**. For example, in a case where there is an article on the conveyance module **140-1**, the presence indicator LED **305** turns on, and in a case where there is no article on the conveyance module **140-1**, the presence indicator LED **305** turns off.

(51) The barcode readers **303** read a barcode attached to an article or a tray. The barcode is attached to identify a conveyance destination, a type of article, and the like.

(52) The ID markers **304** are markers indicating identification information and a direction of the conveyance module **140**. The ID marker **304** is attached to the conveyance module **140-1** in each of the directions A and B.

(53) The light **306** is used for the camera **307** to acquire a clear image. The camera **307** reads the ID marker **304** and a signal of the presence indicator LED **305** of another conveyance module **140**.

(54) The roller **301**, the presence sensor **302**, the barcode reader **303**, the presence indicator LED **305**, and the camera **307** are examples of the conveyance work device group **151**. The camera **307** is an example of the interconnection information acquisition device **153**.

(55) For example, the following basic control logic is set up in the conveyance module **140-1**. In a case where there is no article on the own-conveyance module **140-1**, the conveyance module **140-1** receives an article that is conveyed from an arbitrary direction, and in a case where the presence indicator LED **305** of the conveyance module **140** connected to the conveyance module **140-1** is off, the conveyance module **140-1** conveys the article to the conveyance module **140**.

(56) FIG. **3**B is a perspective view illustrating a specific structure of the conveyance module **140-3** (diverting conveyor). Note that axes illustrated in the perspective view are provided for description.

(57) The conveyance module **140-3** includes rollers **301**, presence sensors **302**, barcode readers **303**, ID markers **304**, a presence indicator LED **305**, lights **306**, and cameras **307**.

(58) The conveyance module **140-3** includes rollers **301** having different rotation directions. Furthermore, the ID marker **304** is attached to the conveyance module **140-3** in each of the directions A, B, C, and D. Furthermore, the conveyance module **140-3** includes the light **306** and the camera **307** in each of the four directions. Other configurations are the same as those of the conveyance module **140-1**.

(59) For example, the following basic control logic is set up in the conveyance module **140-3**. In a case where there is no article on the own-conveyance module **140-3**, the conveyance module **140-3** receives an article that is conveyed from an arbitrary direction, and in a case where the presence indicator LED **305** of the conveyance module **140** connected to the conveyance module **140-3** is off, the conveyance module **140-3** conveys the article to the conveyance module **140**.

(60) FIG. **3**C is a perspective view illustrating a specific structure of a conveyance module **140-4** (picking robot). The conveyance module **140-4** is a robot that performs picking work for articles. Note that axes illustrated in the perspective view are provided for description.
(61) The conveyance module **140-4** includes barcode readers **303**, ID markers **304**, lights **306**, cameras **307**, a hand **310**, a light **311**, an RGBD camera **312**, and a sucker **313**.
(62) The hand **310** moves in various directions for performing picking work. The light **311** is used for the RGBD camera **312** to acquire a clear image. The RGBD camera **312** identifies a good on another conveyance module **140** adjacent to the conveyance module **140-4**. The sucker **313** holds an article.
(63) The ID marker **304** is attached to the conveyance module **140-4** in each of the four directions. Furthermore, the conveyance module **140-4** includes the presence indicator LED **305**, the light **306**, and the camera **307** in each direction. Other configurations are the same as those of the conveyance module **140-1**.
(64) For example, the following basic control logic is set up in the conveyance module **140-4**. In a case where the presence indicator LED **305** of the conveyance module **140** connected to the conveyance module **140-4** is on, the conveyance module **140-4** moves the hand **310** to hold the article.
(65) In the first embodiment, the conveyance module **140** acquires interconnection information by capturing the ID marker **304** by using the camera **307**.
(66) However, the method of acquiring the interconnection information by the conveyance module **140** is an example, and the present invention is not limited thereto. For example, an ID and the like may be notified between the conveyance modules **140** using infrared communication. Furthermore, conveyance module **140** may include an RFID tag and a reader and acquire the interconnection information by reading the RFID tag by the reader. Furthermore, the connection direction may be identified by providing wiring on frames of the conveyance module **140** and connecting or short-circuiting the frames by bringing the frames into contact with each other.
(67) FIG. **4** is a table illustrating an example of a data structure of the conveyance module basic information **121** in the first embodiment.
(68) The conveyance module basic information **121** stores entries formed of type **401**, shape **402**, conveyance control **403**, and ID range **404**. One entry corresponds to one kind of the conveyance module **140**. Note that the fields included in the entry are examples, and the present invention is not limited thereto.
(69) The type **401** is a field that stores values indicating the type of the conveyance module **140**. For example, the model number, the name, and the like of the conveyance module **140** are stored.
(70) The shape **402** is a field that stores values regarding the shape of the conveyance module **140**. For example, values indicating the size and the like of the conveyance module **140** are stored.
(71) The conveyance control **403** is a field that stores values regarding a basic conveyance method of an article. For example, values indicating the conveying direction are stored.
(72) The ID range **404** is a field that stores ranges of the IDs assigned to the same kind of conveyance module **140**.
(73) FIG. **5** is a table illustrating an example of a data structure of the interconnection information **130** in the first embodiment.
(74) The interconnection information **130** is formed of first conveyance module ID **501**, first conveyance module interconnection direction **502**, second conveyance module ID **503**, and second conveyance module interconnection direction **504**.
(75) The first conveyance module ID **501** is a field that stores an ID of a first conveyance module **140**. In the first embodiment, the ID of the own-conveyance module **140** is stored in the first conveyance module ID **501**. Here, the own-conveyance module **140** represents the conveyance module **140** that generates the interconnection information **130**.
(76) The first conveyance module interconnection direction **502** is a field that stores a value

indicating a connection direction of a second conveyance module **140** to the first conveyance module **140**. In the first embodiment, a value indicating the connection direction of another conveyance module **140** to the own-conveyance module **140** is stored in the first conveyance module interconnection direction **502**. Here, the another conveyance module **140** represents the conveyance module **140** connected to the own-conveyance module **140**.
(77) The second conveyance module ID **503** is a field that stores an ID of the second conveyance module **140**. In the first embodiment, the ID of the another conveyance module **140** is stored in the second conveyance module ID **503**.
(78) The second conveyance module interconnection direction **504** is a field that stores a value indicating a connection direction of the first conveyance module **140** to the second conveyance module **140**. In the first embodiment, a value indicating the connection direction of the own-conveyance module **140** to the another conveyance module **140** is stored in the second conveyance module interconnection direction **504**.
(79) In a case where two other conveyance modules **140** are connected to the own-conveyance module **140**, the own-conveyance module **140** generates the interconnection information **130** for each connection and transmits the interconnection information **130** to the integrated control computer **100**.
(80) FIG. **6** is a flowchart describing an example of processing executed by the integrated control computer **100** in the first embodiment. FIG. **7** is a diagram illustrating an example of a conveyor line constructed in the conveyance system **101** in the first embodiment.
(81) The integrated control computer **100** acquires the interconnection information **130** from the conveyance system **101** (step S**101**). In the first embodiment, each conveyance module **140** transmits the interconnection information **130**. The integrated control computer **100** stores the received interconnection information **130** in the memory **111**.
(82) Next, the integrated control computer **100** generates conveyor line model information using the conveyance module basic information **121** and the received interconnection information **130** (step S**102**).
(83) The conveyor line model information is information for managing the constitution of the conveyor line constructed in the conveyance system **101**. The conveyor line model information includes, for example, data indicating a connection relationship of the conveyance modules **140** forming the conveyor line. The integrated control computer **100** can display a diagram as illustrated in FIG. **7** on the basis of the conveyor line model information. Note that the conveyor line model information may not be generated.
(84) In FIG. **7**, bold numerical values on the conveyance modules **140** indicate IDs. Conveyance routes **700** and **701** indicate conveyance routes set up on the conveyor line. Here, the conveyance route **700** is a conveyance route for conveying articles of company A, and the conveyance route **701** is a conveyance route for conveying articles of company B.
(85) Next, the integrated control computer **100** generates conveyance routes (conveyance plan) on the conveyor line (step S**103**).
(86) It is assumed that the integrated control computer **100** has received information about points such as a start point, an end point, and a relay point of the conveyor line, and information regarding the type and the like of the articles conveyed using the conveyor line in advance. The start point, the end point, the relay point, and the like of the conveyor line are designated using identification information and the directions of the conveyance modules **140**. In FIG. **7**, “10015, B”, “10020, A”, and “10017, A” are input as a start point, an end point, and a relay point of the conveyance route **700**, and “10015, B”, “10020, A”, and “10018, A” are input as a start point, an end point, and a relay point of the conveyance route **701**.
(87) The integrated control computer **100** generates conveyance routes connecting the input points on the basis of a known route search algorithm such as Dijkstra's algorithm. Note that the present invention is not limited to the method of searching for conveyance routes. Note that two or more

conveyance routes may be generated on one conveyor line.
(88) Next, the integrated control computer **100** identifies the conveyance module **140** that requires the special control logic to be set up (step S**104**).
(89) For example, the integrated control computer **100** identifies the conveyance module **140** that meets any of the following conditions. (1) The conveyance module **140** has a route set up to convey an article after receiving the article and stopping for a certain period. This is because it is necessary to set up special control logic for conveying the article in a case where predetermined conditions are met after the conveyance module **140** receives the article and stops for a certain period. (2) The conveyance module **140** has two or more routes set up having different conveyance destinations for articles received from the same direction. This is because it is necessary to set up special control logic that controls which article is conveyed at which timing and in which direction.
(90) The conveyance modules **140-1** with ID "10017" and "10018" illustrated in FIG. **7** satisfy condition (1) because they each have a route set up to convey the article after receiving the article and stopping for a certain period. The conveyance modules **140-3** with ID "10033" and "10037" satisfy condition (2) because they each have two routes set up having different conveyance destinations.
(91) Note that the above conditions are examples, and the present invention is not limited thereto. It is sufficient that the conditions can identify the conveyance module **140** that is difficult to control by the basic control logic.
(92) Next, the integrated control computer **100** generates special control logic to be set up in the identified conveyance module **140**, and sets up the special control logic in the conveyance module **140** (step S**105**).
(93) In the case of the conveyance routes illustrated in FIG. **7**, special control logic for performing the following control is generated for the conveyance module **140-1** with ID "10017". (1) Receive an article and stop. (2) In a case where a certain period of time has elapsed or a predetermined condition is satisfied, convey the article in the same direction as the direction in which the conveyance module **140-1** with ID "10017" has received the article.
(94) As in the case of the conveyance routes illustrated in FIG. **7**, special control logic for performing the following control is generated for the conveyance module **140-3** with ID "10033". (1) In a case where the conveyance module **140-3** with ID "10033" has received an article of the company A from a conveyance module **140-3** with ID "10031", convey the article to a conveyance module **140-2** with ID "10021". (2) In a case where the conveyance module **140-3** with ID "10033" has received an article of the company B from the conveyance module **140-3** with ID "10031", convey the article to a conveyance module **140-3** with ID "10035". (3) Do not receive an article of the company A until the conveyance module **140-3** with ID "10033" receives the article from the conveyance module **140-2** with ID "10021". Receive an article of the company B. (4) In a case where the conveyance module **140-3** with ID "10033" has received the article from the conveyance module **140-2** with ID "10021", convey the article to the conveyance module **140-3** with ID "10035".
(95) According to the first embodiment, the conveyance module **140** can transmit the interconnection information **130** indicating the connection relationship between the own-conveyance module **140** and other conveyance modules **140** to the integrated control computer **100**. As a result, the integrated control computer **100** can grasp the conveyor line constructed in the conveyance system **101**. Furthermore, the integrated control computer **100** can automatically perform generation of conveyor line model information, generation of a conveyance route, setting up of special control logic, and the like on the basis of the interconnection information **130** received from the conveyance system **101**.
Second Embodiment
(96) In the second embodiment, a conveyance vehicle that autonomously travels is included as the conveyance module **140**. Hereinafter, the second embodiment will be described focusing on a

difference from the first embodiment.
(97) The configuration of a system in the second embodiment is the same as the configuration in the first embodiment. The configuration of the integrated control computer **100** in the second embodiment is the same as the configuration in the first embodiment. The configuration of the conveyance system **101** in the second embodiment is the same as that in the first embodiment, but a conveyance vehicle is included as a conveyance module **140-5**. The conveyance vehicle receives an article, autonomously travels to move to a target conveyance module **140**, and conveys the article to the conveyance module **140**.
(98) The data structure of the conveyance module basic information **121** in the second embodiment is the same as the data structure in the first embodiment. The data structure of the interconnection information **130** in the second embodiment is the same as the data structure in the first embodiment.
(99) The conveyance modules **140-1**, **140-2**, **140-3**, and **140-4** in the second embodiment do not include the interconnection information acquisition device **153**. On the other hand, the conveyance vehicle (conveyance module **140-5**) includes the interconnection information acquisition device **153**.
(100) Structures of the conveyance modules **140-1**, **140-2**, **140-3**, and **140-4** in the second embodiment are the same as those structures of the conveyance modules in the first embodiment. However, in the second embodiment, the camera **307** is not used for reading the ID marker **304**.
(101) FIG. **8** is a diagram illustrating a specific structure of the conveyance module **140-5** in the second embodiment.
(102) The conveyance module **140-5** includes rollers **301**, presence sensors **302**, barcode readers **303**, ID markers **304**, presence indicator LEDs **305**, lights **306**, and cameras **307**. Furthermore, the conveyance module **140-5** includes a distance sensor and a tire for autonomous traveling (not illustrated).
(103) The conveyance module **140-5** includes rollers **301** having different rotation directions. Furthermore, the ID marker **304** is attached to the conveyance module **140-5** in each of the directions A, B, C, and D. Furthermore, the conveyance module **140-5** includes the presence indicator LED **305**, the light **306**, and the camera **307** in each of the four directions. Other configurations are the same as those of the conveyance module **140-1**.
(104) The controller **150** of the conveyance module **140-5** stores map information of the movement space. The conveyance module **140-5** travels in the movement space while estimating its own position on the basis of the map information and a measurement result of the distance sensor. The conveyance module **140-5** may randomly travel in the movement space, or may travel in the movement space on the basis of a preset policy.
(105) The conveyance module **140-5** uses the cameras **307** to read the ID markers **304** attached to the conveyance modules **140** forming the conveyor line while traveling, and generates the interconnection information **130**. At this time of operation, the conveyance module **140-5** determines one of the conveyance modules **140** connected as a first conveyance module. For example, the first conveyance module is determined according to the traveling direction. The conveyance module **140-5** transmits the interconnection information **130** to the integrated control computer **100**.
(106) The processing executed by the integrated control computer **100** in the second embodiment is the same as that in the first embodiment. However, in the second embodiment, a large-scale conveyor line formed of independent conveyor lines can be constructed by using the conveyance module **140-5**. The conveyance of the article between the independent conveyor lines is performed by the conveyance module **140-5**.
(107) FIG. **9** is a diagram illustrating an example of a conveyor line constructed in the conveyance system **101** in the second embodiment.
(108) The conveyor line illustrated in FIG. **9** is formed of four conveyor lines **900-1**, **900-2**, **900-3**,

and **900-4**. Note that the conveyor line **900-3** is a conveyor line constructed by one conveyance module **140-1**.

(109) The conveyor lines **900-1**, **900-2**, and **900-4** include robots **910** that are movable along rails of pedestals **911**. The robot **910** takes out cases **921** from a shelf **920**, or accommodates the cases **921** in the shelf **920**. The conveyance module **140-5** conveys the cases **921** between the conveyor lines **900-1**, **900-2**, and **900-3**. The conveyance module **140-5** is not treated as the conveyance module **140** forming a conveyor line.

(110) The conveyor line illustrated in FIG. **9** is a conveyor line constructed for performing a sorting work of articles. For example, in the conveyor lines **900-1** and **900-2**, the cases **921** with articles accommodated therein are taken out from the shelves **920**, and the cases **921** are conveyed to the conveyor line **900-3** by the conveyance module **140-5**. The conveyance module **140-4** connected to the conveyance module **140-1** of the conveyor line **900-3** distributes the cases **921** to conveyance routes set up on the conveyor line, and the robots **910** hold the cases **921** that have arrived at end points of the conveyance routes and accommodate the cases **921** in the shelves **920**.

(111) Note that control logic is set up in each robot **910** in advance. Furthermore, special control logic is set up in the conveyance module **140-5**.

(112) FIG. **10** is a diagram illustrating an example of a conveyor line constructed in the conveyance system **101** in the second embodiment.

(113) The conveyor line illustrated in FIG. **10** is formed of four conveyor lines **1000-1**, **1000-2**, **1000-3**, and **1000-4**. Note that the conveyor line **1000-3** is a conveyor line constructed by one conveyance module **140-1**.

(114) The conveyor lines **1000-1**, **1000-2**, and **1000-4** include manufacturing robots **1010**. The robot **1010** is a robot that manufactures articles such as processing, welding, and assembly of articles **1011**. The conveyance module **140-5** conveys the manufactured articles between the conveyor lines **1000-1**, **1000-2**, and **1000-3**. The conveyance module **140-5** is not treated as the conveyance module **140** forming a conveyor line.

(115) The conveyor line illustrated in FIG. **10** is a conveyor line constructed for performing a manufacturing operation of articles. For example, in the conveyor lines **1000-1** and **1000-2**, work such as processing of the articles **1011** is performed, and the articles are conveyed to the conveyor line **1000-3** by the conveyance module **140-5**. The conveyance module **140-4** connected to the conveyance module **140-1** of the conveyor line **1000-3** distributes the articles to conveyance routes set up on the conveyor line, and the robots **1010** perform work such as processing on the articles that have arrived at end points of the conveyance routes.

(116) Note that control logic is set up in each robot **1010** in advance. Furthermore, special control logic is set up in the conveyance module **140-5**.

(117) According to the second embodiment, the conveyance module **140-5** can acquire interconnection information **130** by traveling in the movement space in the conveyance system **101**, and can transmit the interconnection information **130** to the integrated control computer **100**. As a result, the integrated control computer **100** can grasp the conveyor line constructed in the conveyance system **101**. Furthermore, the integrated control computer **100** can automatically perform generation of conveyor line model information, generation of a conveyance route, setting up of special control logic, and the like on the basis of the interconnection information **130** received from the conveyance system **101**.

(118) Note that the present invention is not limited to the above embodiments, and includes various modifications. Furthermore, for example, the above embodiments are detailed descriptions of the configurations for the purpose of making the present invention easier to understand, and are not necessarily limited to those having all the described configurations. Furthermore, a part of the configuration of each embodiment can be added to, deleted from, or replaced with another configuration.

(119) Furthermore, some or all of the configurations, functions, processing units, processing means,

and the like in each configuration described above may be realized by hardware, for example, by designing with an integrated circuit or the like. Furthermore, the present invention can also be realized by a program code of software that realizes the functions of the embodiments. In this case, a storage medium in which a program code is recorded is provided to a computer, and a processor included in the computer reads the program code stored in the storage medium. In this case, the program code itself read from the storage medium realizes the functions of the above-described embodiments, and therefore the program code itself and the storage medium in which the program code is stored form the present invention. As a storage medium for supplying such a program code, for example, a flexible disk, a CD-ROM, a DVD-ROM, a hard disk, a solid state drive (SSD), an optical disk, a magneto-optical disk, a CD-R, a magnetic tape, a non-volatile memory card, a ROM, or the like is used.
(120) Furthermore, the program code that realizes the functions described in the present embodiments can be implemented by, for example, an assembler or a wide range of programs or script languages such as C/C++, perl, Shell, PHP, Python, or Java.
(121) Moreover, the program code of software that realizes the functions of the embodiments is delivered via a network so that the program code may be stored in a storage means such as a hard disk or a memory of a computer or a storage medium such as a CD-RW or a CD-R, and the processor included in the computer may read and execute the program code stored in the storage means or the storage medium.
(122) In the above embodiments, control lines and information lines considered to be necessary for description are illustrated, but all the control lines and the information lines are not necessarily illustrated in products. All the constituents may be connected to each other.

Claims

1. A computer system comprising a conveyance system, and a control system communicably connected to the conveyance system, wherein the conveyance system includes a plurality of conveyance modules each having a conveyance means for conveying an article, the control system includes at least one computer having an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, the conveyance system transmits, to the control system, interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line, and the control system controls the conveyance system on a basis of a plurality of pieces of the interconnection information, wherein the interconnection information includes identification information of a first conveyance module, identification information of a second conveyance module connected to the first conveyance module, a connection direction of the second conveyance module to the first conveyance module, and a connection direction of the first conveyance module to the second conveyance module, and wherein the control system holds conveyance module basic information for managing a structure of the plurality of conveyance modules, and generates conveyor line model information for managing a structure of the conveyor line constructed in the conveyance system using the conveyance module basic information and the plurality of pieces of interconnection information.
2. A computer system comprising a conveyance system, and a control system communicably connected to the conveyance system, wherein the conveyance system includes a plurality of conveyance modules each having a conveyance means for conveying an article, the control system includes at least one computer having an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, the conveyance system transmits, to the control system, interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line, and the control system controls the conveyance system on a basis of a plurality of pieces of the

interconnection information, wherein the interconnection information includes identification information of a first conveyance module, identification information of a second conveyance module connected to the first conveyance module, a connection direction of the second conveyance module to the first conveyance module, and a connection direction of the first conveyance module to the second conveyance module, and wherein the control system holds conveyance module basic information for managing a structure of the plurality of conveyance modules, and generates a conveyance route in the conveyor line constructed in the conveyance system using the conveyance module basic information and the plurality of pieces of interconnection information.

3. The computer system according to claim 2, wherein basic control logic that defines a basic control method of the conveyance means is set up in each of the plurality of conveyance modules, the control system identifies at least one of the conveyance module that requires additional control logic to be set up in addition to the basic control logic on a basis of the conveyance route among the plurality of conveyance modules, generates special control logic to be set up in the conveyance module identified, and sets up the special control logic in the identified conveyance module.

4. A computer system comprising a conveyance system, and a control system communicably connected to the conveyance system, wherein the conveyance system includes a plurality of conveyance modules each having a conveyance means for conveying an article, the control system includes at least one computer having an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, the conveyance system transmits, to the control system, interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line, and the control system controls the conveyance system on a basis of a plurality of pieces of the interconnection information, wherein the interconnection information includes identification information of a first conveyance module, identification information of a second conveyance module connected to the first conveyance module, a connection direction of the second conveyance module to the first conveyance module, and a connection direction of the first conveyance module to the second conveyance module, and wherein each of the plurality of conveyance modules includes an acquisition means that acquires the interconnection information, acquires the interconnection information in which the first conveyance module is set as an own-conveyance module using the acquisition means, and transmits the interconnection information to the control system.

5. A computer system comprising a conveyance system, and a control system communicably connected to the conveyance system, wherein the conveyance system includes a plurality of conveyance modules each having a conveyance means for conveying an article, the control system includes at least one computer having an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, the conveyance system transmits, to the control system, interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line, and the control system controls the conveyance system on a basis of a plurality of pieces of the interconnection information, wherein the interconnection information includes identification information of a first conveyance module, identification information of a second conveyance module connected to the first conveyance module, a connection direction of the second conveyance module to the first conveyance module, and a connection direction of the first conveyance module to the second conveyance module, and wherein the plurality of conveyance modules include a moving conveyance module that can autonomously travel between the conveyance modules forming the conveyor line for conveying the article, the moving conveyance module includes an acquisition means that acquires the interconnection information, travels in a space in which the plurality of conveyance modules are arranged and acquires the interconnection information of each of the plurality of conveyance modules using the acquisition means, and transmits the interconnection information of each of the plurality of conveyance modules to the control system.

6. A computer communicably connected to a conveyance system including a plurality of conveyance modules each having a conveyance means for conveying an article, the computer comprising an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, wherein the arithmetic device acquires interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line from the conveyance system via the communication device, and controls the conveyance system on a basis of a plurality of pieces of the interconnection information, wherein the interconnection information includes identification information of a first conveyance module, identification information of a second conveyance module connected to the first conveyance module, a connection direction of the second conveyance module to the first conveyance module, and a connection direction of the first conveyance module to the second conveyance module, and wherein the storage device stores conveyance module basic information for managing a structure of the plurality of conveyance modules, and the arithmetic device generates conveyor line model information for managing a structure of the conveyor line constructed in the conveyance system using the conveyance module basic information and the plurality of pieces of interconnection information, and stores the conveyor line model information in the storage device.
7. A computer communicably connected to a conveyance system including a plurality of conveyance modules each having a conveyance means for conveying an article, the computer comprising an arithmetic device, a storage device connected to the arithmetic device, and a communication device connected to the arithmetic device, wherein the arithmetic device acquires interconnection information regarding connection between the conveyance modules of each of the plurality of conveyance modules forming a conveyor line from the conveyance system via the communication device, and controls the conveyance system on a basis of a plurality of pieces of the interconnection information, wherein the interconnection information includes identification information of a first conveyance module, identification information of a second conveyance module connected to the first conveyance module, a connection direction of the second conveyance module to the first conveyance module, and a connection direction of the first conveyance module to the second conveyance module, and wherein the storage device stores conveyance module basic information for managing a structure of the plurality of conveyance modules, and the arithmetic device generates a conveyance route in the conveyor line constructed in the conveyance system using the conveyance module basic information and the plurality of pieces of interconnection information, and stores information about the conveyance route in the storage device.
8. The computer according to claim 7, wherein basic control logic that is a basic control method of the conveyance means is set up in each of the plurality of conveyance modules, the arithmetic device identifies the conveyance module that requires additional control logic to be set up in addition to the basic control logic on a basis of the information about the conveyance route, generates special control logic set up in the conveyance module identified, and sets up the special control logic in the identified conveyance module.